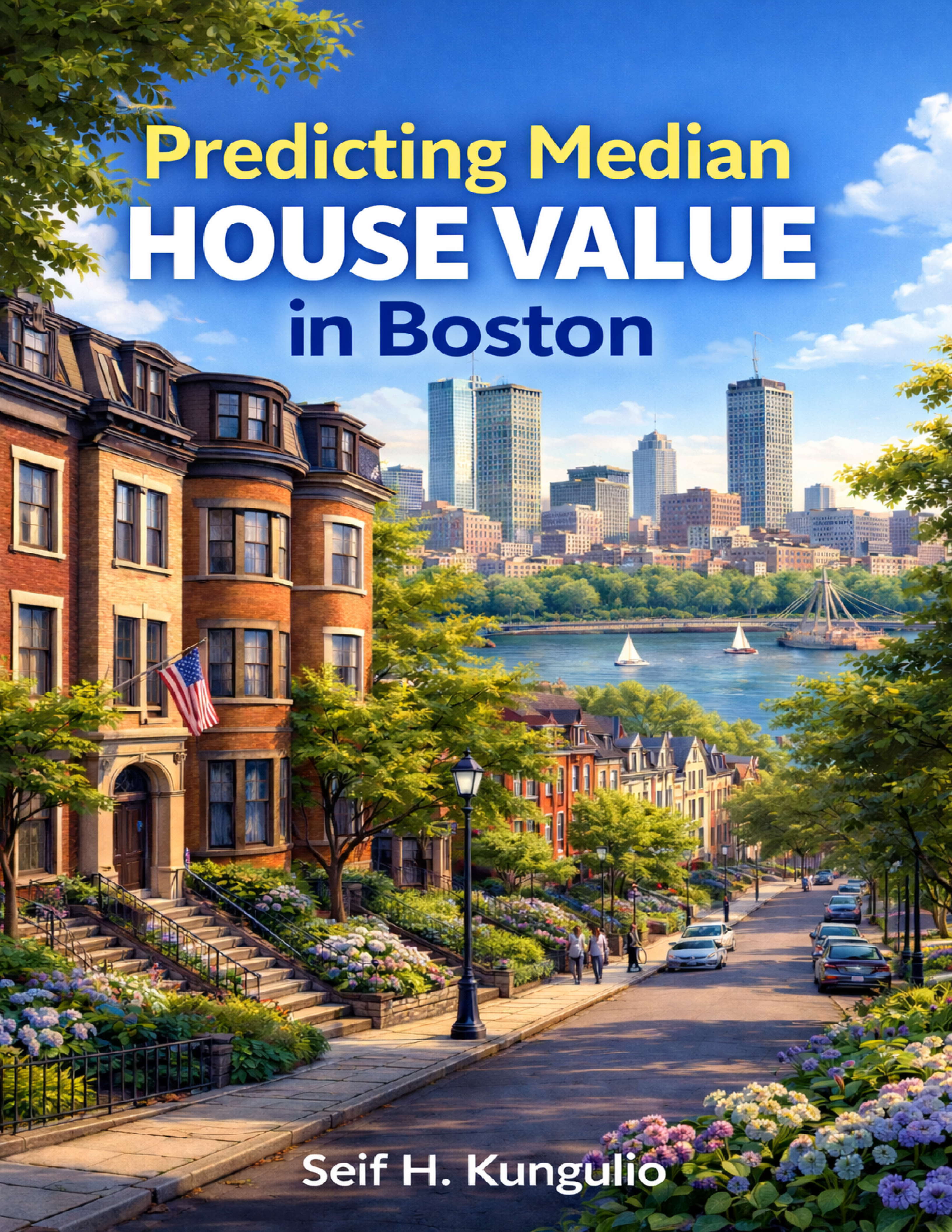
Predicting Median HOUSE VALUE in Boston



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Business Understanding

Background & Problem Statement

Boston's housing authority and urban developers face the challenge of accurately estimating property values across different neighborhoods. Property valuation depends on multiple interrelated factors—such as crime rate, accessibility to major roads, environmental quality, and local amenities—which can be difficult to model using traditional statistical techniques alone.

Business Objectives

Build a predictive model to estimate median owner-occupied home value (medv) using neighborhood-level indicators. The goal is to support:

- neighborhood investment planning,
- redevelopment prioritization,
- and value forecasting under changing conditions.

Success Criteria

- **Primary metric:** RMSE (lower is better)
- **Secondary metrics:** MAE and R^2
- **Practical goal:** A model that generalizes well (low validation error) and offers interpretable drivers (feature importance / coefficients).

Data Understanding

Data Collection

The dataset used in this analysis is the Boston Housing Dataset, which contains information about various attributes of houses in Boston suburbs. The dataset is publicly available and can be accessed through the `MASS` package in R.

```r
data("Boston", package = "MASS")
boston.df <- Boston

glimpse(boston.df)
```

```
## Rows: 506
## Columns: 14
## $ crim    <dbl> 0.00632, 0.02731, 0.02729, 0.03237, 0.06905, 0.02985, 0.08829,~
## $ zn      <dbl> 18.0, 0.0, 0.0, 0.0, 0.0, 0.0, 12.5, 12.5, 12.5, 12.5, 12.5, 1~
## $ indus   <dbl> 2.31, 7.07, 7.07, 2.18, 2.18, 2.18, 7.87, 7.87, 7.87, 7.87, 7.~
## $ chas    <int> 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,~
## $ nox     <dbl> 0.538, 0.469, 0.469, 0.458, 0.458, 0.458, 0.524, 0.524, 0.524,~
## $ rm      <dbl> 6.575, 6.421, 7.185, 6.998, 7.147, 6.430, 6.012, 6.172, 5.631,~
## $ age     <dbl> 65.2, 78.9, 61.1, 45.8, 54.2, 58.7, 66.6, 96.1, 100.0, 85.9, 9~
## $ dis     <dbl> 4.0900, 4.9671, 4.9671, 6.0622, 6.0622, 6.0622, 5.5605, 5.9505~
## $ rad     <int> 1, 2, 2, 3, 3, 3, 5, 5, 5, 5, 5, 5, 5, 4, 4, 4, 4, 4, 4, 4, 4,~
## $ tax     <dbl> 296, 242, 242, 222, 222, 222, 311, 311, 311, 311, 311, 311, 31~
## $ ptratio <dbl> 15.3, 17.8, 17.8, 18.7, 18.7, 18.7, 15.2, 15.2, 15.2, 15.2, 15~
## $ black   <dbl> 396.90, 396.90, 392.83, 394.63, 396.90, 394.12, 395.60, 396.90~
## $ lstat   <dbl> 4.98, 9.14, 4.03, 2.94, 5.33, 5.21, 12.43, 19.15, 29.93, 17.10~
## $ medv    <dbl> 24.0, 21.6, 34.7, 33.4, 36.2, 28.7, 22.9, 27.1, 16.5, 18.9, 15~
```

```r
summary(boston.df$medv)
```

```
##    Min. 1st Qu.  Median    Mean 3rd Qu.    Max. 
##    5.00   17.02   21.20   22.53   25.00   50.00
```

Data Description

The dataset used in this project is the Boston Housing Dataset, originally published in the Harrison and Rubinfeld (1978) study and included in the MASS package in R. It contains 506 observations and 14 variables describing socioeconomic, demographic, environmental, and structural characteristics of housing in Boston suburbs. The dataset is commonly used for predictive modeling and regression analysis tasks, particularly for estimating housing values.

Each record represents a census tract in the Boston metropolitan area, with the target variable being the median value of owner-occupied homes (MEDV) expressed in $1000s.

The features cover a wide range of factors that influence property values — including crime rates, industrial activity, environmental quality, accessibility to major roads, and education levels. These variables provide a multi-dimensional view of the housing environment, making the dataset suitable for developing regression and machine learning models to predict housing prices.

Data Dictionary

Variable	Description	Data Type	Constraints/Rule
crim	Per capita crime rate by town	Numeric	>= 0
zn	Proportion of residential land zoned for lots over 25,000 sq.ft.	Numeric	0 <= **zn** <= 100
indus	Proportion of non-retail business acres per town	Numeric	>= 0
chas	Charles River dummy variable (1 if tract bounds river; 0 otherwise)	Integer	0 or 1
nox	Nitric oxides concentration (parts per 10 million)	Numeric	0 < **nox** <= 1
rm	Average number of rooms per dwelling	Numeric	> 0
age	Proportion of owner-occupied units built prior to 1940 (%)	Numeric	0 <= **age** <= 100
dis	Weighted distances to five Boston employment centers	Numeric	> 0
rad	Index of accessibility to radial highways	Integer	1–24
tax	Full-value property-tax rate per $10,000	Numeric	> 0
ptratio	Pupil-teacher ratio by town	Numeric	> 0
black	(1000(Bk - 0.63)^2), where Bk is the proportion of Black residents by town	Numeric	>= 0
lstat	Percentage of lower status of the population	Numeric	0 <= **lstat** <= 100
medv	Median value of owner-occupied homes in $1000s (Target variable)	Numeric	> 0 (typically capped at 50 in dataset)

Data Quality Assessment

```
# Missing values
colSums(is.na(boston.df))
```

```
##    crim      zn   indus    chas     nox      rm     age     dis     rad     tax
##       0       0       0       0       0       0       0       0       0       0
## ptratio   black   lstat    medv
##       0       0       0       0
```

```
# Duplicates
sum(duplicated(boston.df))
```

```
## [1] 0
```

```
# Quick profile
skim(boston.df)
```

Table 2: Data summary

Name	boston.df
Number of rows	506

Number of columns	14
Column type frequency:	
numeric	14
Group variables	None

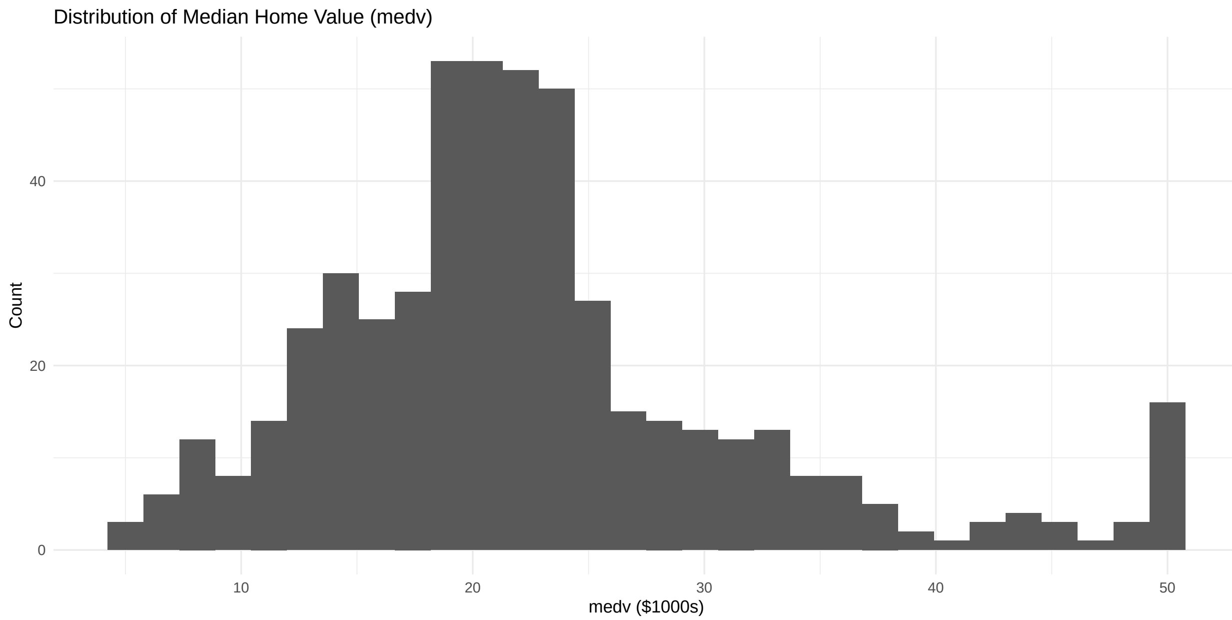
Variable type: numeric

skim_variable	n_missing	complete_rate	mean	sd	p0	p25	p50	p75	p100	hist
crim	0	1	3.61	8.60	0.01	0.08	0.26	3.68	88.98	
zn	0	1	11.36	23.32	0.00	0.00	0.00	12.50	100.00	
indus	0	1	11.14	6.86	0.46	5.19	9.69	18.10	27.74	
chas	0	1	0.07	0.25	0.00	0.00	0.00	0.00	1.00	
nox	0	1	0.55	0.12	0.38	0.45	0.54	0.62	0.87	
rm	0	1	6.28	0.70	3.56	5.89	6.21	6.62	8.78	
age	0	1	68.57	28.15	2.90	45.02	77.50	94.07	100.00	
dis	0	1	3.80	2.11	1.13	2.10	3.21	5.19	12.13	
rad	0	1	9.55	8.71	1.00	4.00	5.00	24.00	24.00	
tax	0	1	408.24	168.54	187.00	279.00	330.00	666.00	711.00	
ptratio	0	1	18.46	2.16	12.60	17.40	19.05	20.20	22.00	
black	0	1	356.67	91.29	0.32	375.38	391.44	396.22	396.90	
lstat	0	1	12.65	7.14	1.73	6.95	11.36	16.96	37.97	
medv	0	1	22.53	9.20	5.00	17.02	21.20	25.00	50.00	

Exploratory Data Analysis (EDA)

Target variable distribution

```r
ggplot(boston.df, aes(medv)) +
  geom_histogram(bins = 30) +
  labs(title = "Distribution of Median Home Value (medv)",
       x = "medv ($1000s)",
       y = "Count")
```

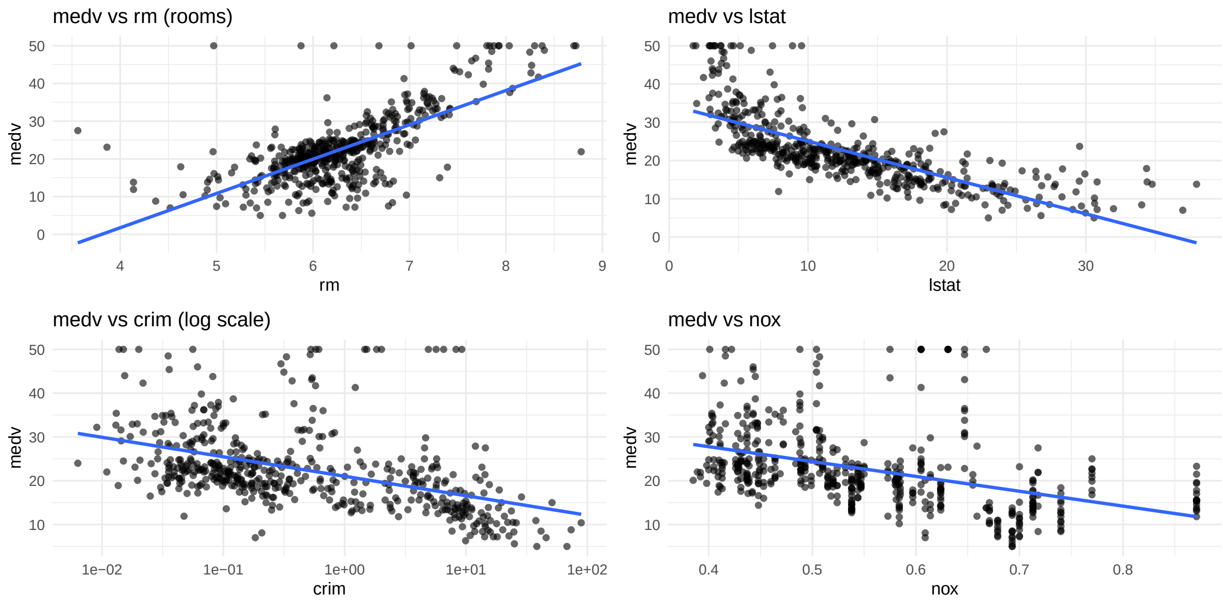


Relationship with key predictors

```
p1 <- ggplot(boston.df, aes(rm, medv)) + geom_point(alpha = 0.6) +
  geom_smooth(method="lm", se=FALSE) +
  labs(title="medv vs rm (rooms)")
p2 <- ggplot(boston.df, aes(lstat, medv)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method="lm", se=FALSE) +
  labs(title="medv vs lstat")
p3 <- ggplot(boston.df, aes(crim, medv)) +
  geom_point(alpha = 0.6) + scale_x_log10() +
  geom_smooth(method="lm", se=FALSE) +
  labs(title="medv vs crim (log scale)")
p4 <- ggplot(boston.df, aes(nox, medv)) +
  geom_point(alpha = 0.6) +
  geom_smooth(method="lm", se=FALSE) +
  labs(title="medv vs nox")

(p1 | p2) / (p3 | p4)
```

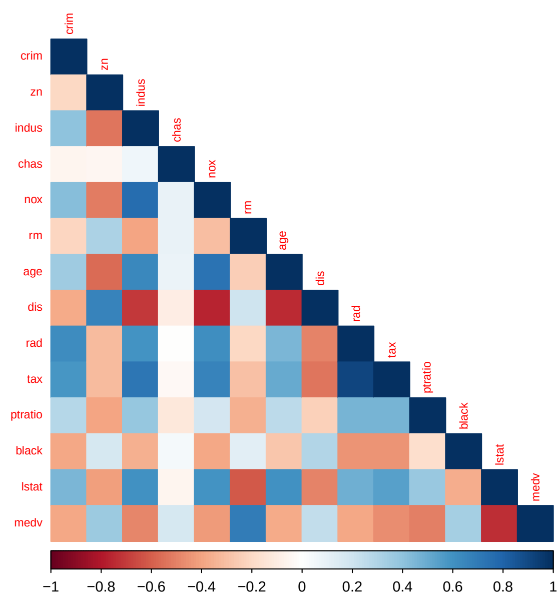
```
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
## `geom_smooth()` using formula = 'y ~ x'
```



Correlation Overview

```r
num_df <- boston.df %>% select(where(is.numeric))
corr <- cor(num_df)

corrplot(corr, method = "color", type = "lower", tl.cex = 0.7)
```



Data Preparation

Train/ Test Split

```
set.seed(123)
idx <- createDataPartition(boston.df$medv, p = 0.8, list = FALSE)
train <- boston.df[idx, ]
test  <- boston.df[-idx, ]

dim(train); dim(test)
```

```
## [1] 407  14
```

```
## [1] 99 14
```

Preprocessing Recipe

```
preproc <- preProcess(
  train %>% select(-medv),
  method = c("center", "scale")
)

x_train <- predict(preproc, train %>% select(-medv))
x_test  <- predict(preproc, test %>% select(-medv))

y_train <- train$medv
y_test  <- test$medv
```

Modeling

Evaluation Setup (Cross-Validation)

```
ctrl <- trainControl(
  method = "repeatedcv",
  number = 10,
  repeats = 3
)

rmse <- function(actual, pred) sqrt(mean((actual - pred)^2))
mae  <- function(actual, pred) mean(abs(actual - pred))
r2   <- function(actual, pred) cor(actual, pred)^2
```

Baseline Model (Mean Predictor)

```
baseline_pred <- rep(mean(y_train), length(y_test))

baseline_metrics <- tibble(
  Model = "Baseline (Train Mean)",
  RMSE = rmse(y_test, baseline_pred),
  MAE  = mae(y_test, baseline_pred),
  R2   = r2(y_test, baseline_pred)
)
```

```
## Warning in cor(actual, pred): the standard deviation is zero
```

```
baseline_metrics
```

```
## # A tibble: 1 x 4
##   Model                  RMSE   MAE    R2
##   <chr>                 <dbl> <dbl> <dbl>
## 1 Baseline (Train Mean)  9.32  6.60    NA
```

Multiple Linear Regression

```
lm_fit <- train(
  medv ~ ., data = train,
  method = "lm",
  trControl = ctrl
)

lm_pred <- predict(lm_fit, newdata = test)

lm_metrics <- tibble(
  Model = "Linear Regression",
```

```r
  RMSE = rmse(y_test, lm_pred),
  MAE  = mae(y_test, lm_pred),
  R2   = r2(y_test, lm_pred)
)

lm_fit
```

```
## Linear Regression
##
## 407 samples
##  13 predictor
##
## No pre-processing
## Resampling: Cross-Validated (10 fold, repeated 3 times)
## Summary of sample sizes: 367, 366, 366, 367, 366, 366, ...
## Resampling results:
##
##   RMSE      Rsquared   MAE
##   4.867445  0.7264844  3.42562
##
## Tuning parameter 'intercept' was held constant at a value of TRUE
```

```r
lm_metrics
```

```
## # A tibble: 1 x 4
##   Model              RMSE   MAE    R2
##   <chr>             <dbl> <dbl> <dbl>
## 1 Linear Regression  4.59  3.37 0.761
```

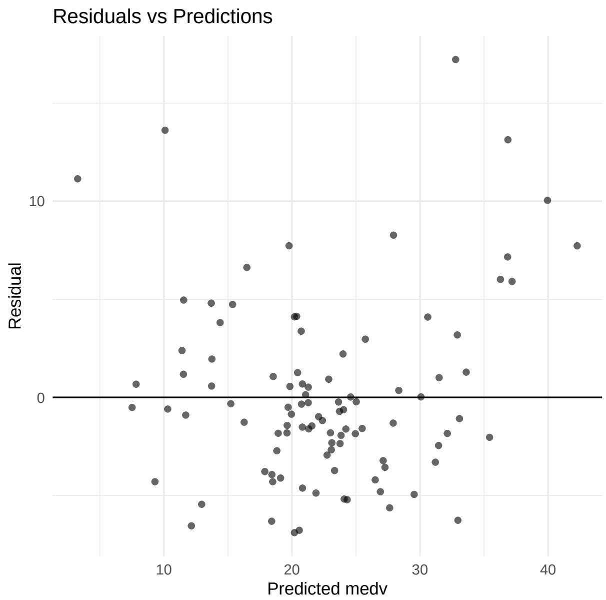
Diagnostics (Residuals)

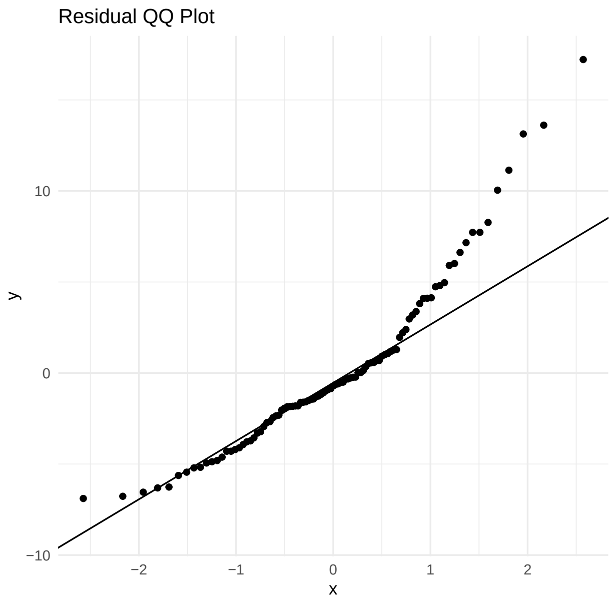
```r
res <- y_test - lm_pred

p_res1 <- ggplot(tibble(pred=lm_pred, res=res), aes(pred, res)) +
  geom_point(alpha=0.6) + geom_hline(yintercept = 0) +
  labs(title="Residuals vs Predictions", x="Predicted medv", y="Residual")

p_res2 <- ggplot(tibble(res=res), aes(sample=res)) +
  stat_qq() + stat_qq_line() +
  labs(title="Residual QQ Plot")

p_res1 | p_res2
```





Regularized Regression (Elastic Net via glmnet)

```
grid_glmnet <- expand.grid(
  alpha  = seq(0, 1, by = 0.25),
  lambda = 10^seq(-3, 1, length.out = 40)
)

glmnet_fit <- train(
  x = as.matrix(x_train),
  y = y_train,
  method = "glmnet",
  trControl = ctrl,
  tuneGrid = grid_glmnet
)
```

```
## Warning in nominalTrainWorkflow(x = x, y = y, wts = weights, info = trainInfo,
## : There were missing values in resampled performance measures.
```

```
glmnet_pred <- predict(glmnet_fit, newdata = as.matrix(x_test))

glmnet_metrics <- tibble(
  Model = "Elastic Net (glmnet)",
  RMSE = rmse(y_test, glmnet_pred),
  MAE  = mae(y_test, glmnet_pred),
  R2   = r2(y_test, glmnet_pred)
)

glmnet_fit
```

```
## glmnet
```

```
##
## 407 samples
##  13 predictor
##
## No pre-processing
## Resampling: Cross-Validated (10 fold, repeated 3 times)
## Summary of sample sizes: 366, 367, 366, 367, 365, 367, ...
## Resampling results across tuning parameters:
##
##   alpha  lambda        RMSE      Rsquared   MAE
##   0.00    0.001000000  4.832917  0.7277185  3.355326
##   0.00    0.001266380  4.832917  0.7277185  3.355326
##   0.00    0.001603719  4.832917  0.7277185  3.355326
##   0.00    0.002030918  4.832917  0.7277185  3.355326
##   0.00    0.002571914  4.832917  0.7277185  3.355326
##   0.00    0.003257021  4.832917  0.7277185  3.355326
##   0.00    0.004124626  4.832917  0.7277185  3.355326
##   0.00    0.005223345  4.832917  0.7277185  3.355326
##   0.00    0.006614741  4.832917  0.7277185  3.355326
##   0.00    0.008376776  4.832917  0.7277185  3.355326
##   0.00    0.010608184  4.832917  0.7277185  3.355326
##   0.00    0.013433993  4.832917  0.7277185  3.355326
##   0.00    0.017012543  4.832917  0.7277185  3.355326
##   0.00    0.021544347  4.832917  0.7277185  3.355326
##   0.00    0.027283334  4.832917  0.7277185  3.355326
##   0.00    0.034551073  4.832917  0.7277185  3.355326
##   0.00    0.043754794  4.832917  0.7277185  3.355326
##   0.00    0.055410203  4.832917  0.7277185  3.355326
##   0.00    0.070170383  4.832917  0.7277185  3.355326
##   0.00    0.088862382  4.832917  0.7277185  3.355326
##   0.00    0.112533558  4.832917  0.7277185  3.355326
##   0.00    0.142510267  4.832917  0.7277185  3.355326
##   0.00    0.180472177  4.832917  0.7277185  3.355326
##   0.00    0.228546386  4.832917  0.7277185  3.355326
##   0.00    0.289426612  4.832917  0.7277185  3.355326
##   0.00    0.366524124  4.832917  0.7277185  3.355326
##   0.00    0.464158883  4.832917  0.7277185  3.355326
##   0.00    0.587801607  4.832917  0.7277185  3.355326
##   0.00    0.744380301  4.836849  0.7274231  3.355220
##   0.00    0.942668455  4.847351  0.7265883  3.356558
##   0.00    1.193776642  4.863034  0.7254231  3.362947
##   0.00    1.511775071  4.885406  0.7238610  3.372637
##   0.00    1.914481976  4.916719  0.7217748  3.387967
##   0.00    2.424462017  4.959320  0.7190457  3.412150
##   0.00    3.070290630  5.016077  0.7154868  3.447561
##   0.00    3.888155180  5.089893  0.7109021  3.497191
##   0.00    4.923882632  5.183249  0.7050648  3.563397
##   0.00    6.235507341  5.298286  0.6977548  3.648744
##   0.00    7.896522868  5.435376  0.6888018  3.754453
##   0.00   10.000000000  5.594498  0.6781015  3.881450
##   0.25    0.001000000  4.836438  0.7279316  3.421037
##   0.25    0.001266380  4.836438  0.7279316  3.421037
##   0.25    0.001603719  4.836438  0.7279316  3.421037
##   0.25    0.002030918  4.836438  0.7279316  3.421037
```

```
##    0.25    0.002571914  4.836438  0.7279316  3.421037
##    0.25    0.003257021  4.836438  0.7279316  3.421037
##    0.25    0.004124626  4.836438  0.7279316  3.421037
##    0.25    0.005223345  4.836438  0.7279316  3.421037
##    0.25    0.006614741  4.836438  0.7279316  3.421037
##    0.25    0.008376776  4.836438  0.7279316  3.421037
##    0.25    0.010608184  4.836438  0.7279316  3.421037
##    0.25    0.013433993  4.836438  0.7279316  3.421037
##    0.25    0.017012543  4.836126  0.7279642  3.420499
##    0.25    0.021544347  4.835448  0.7280174  3.418529
##    0.25    0.027283334  4.834691  0.7280719  3.416190
##    0.25    0.034551073  4.833752  0.7281342  3.413073
##    0.25    0.043754794  4.832779  0.7281915  3.409349
##    0.25    0.055410203  4.831802  0.7282406  3.405026
##    0.25    0.070170383  4.831070  0.7282552  3.400320
##    0.25    0.088862382  4.830387  0.7282608  3.394986
##    0.25    0.112533558  4.830098  0.7282192  3.389107
##    0.25    0.142510267  4.830765  0.7280725  3.382744
##    0.25    0.180472177  4.833037  0.7277600  3.376673
##    0.25    0.228546386  4.837666  0.7271960  3.371650
##    0.25    0.289426612  4.846338  0.7262152  3.368771
##    0.25    0.366524124  4.859747  0.7247570  3.370375
##    0.25    0.464158883  4.882315  0.7223572  3.382206
##    0.25    0.587801607  4.914495  0.7189612  3.405712
##    0.25    0.744380301  4.947600  0.7156659  3.431511
##    0.25    0.942668455  4.983606  0.7125137  3.459282
##    0.25    1.193776642  5.032040  0.7085892  3.493198
##    0.25    1.511775071  5.102265  0.7029653  3.545636
##    0.25    1.914481976  5.193345  0.6959931  3.615083
##    0.25    2.424462017  5.270615  0.6934212  3.673924
##    0.25    3.070290630  5.370804  0.6918821  3.753471
##    0.25    3.888155180  5.512105  0.6903937  3.869596
##    0.25    4.923882632  5.707120  0.6887406  4.020335
##    0.25    6.235507341  5.978255  0.6860340  4.221638
##    0.25    7.896522868  6.332114  0.6827414  4.480222
##    0.25   10.000000000  6.773510  0.6804070  4.802855
##    0.50    0.001000000  4.836713  0.7279107  3.421480
##    0.50    0.001266380  4.836713  0.7279107  3.421480
##    0.50    0.001603719  4.836713  0.7279107  3.421480
##    0.50    0.002030918  4.836713  0.7279107  3.421480
##    0.50    0.002571914  4.836713  0.7279107  3.421480
##    0.50    0.003257021  4.836713  0.7279107  3.421480
##    0.50    0.004124626  4.836713  0.7279107  3.421480
##    0.50    0.005223345  4.836713  0.7279107  3.421480
##    0.50    0.006614741  4.836713  0.7279107  3.421480
##    0.50    0.008376776  4.836713  0.7279107  3.421480
##    0.50    0.010608184  4.836697  0.7279150  3.421381
##    0.50    0.013433993  4.836031  0.7279773  3.419769
##    0.50    0.017012543  4.835262  0.7280360  3.417458
##    0.50    0.021544347  4.834449  0.7280919  3.414616
##    0.50    0.027283334  4.833635  0.7281389  3.411206
##    0.50    0.034551073  4.832891  0.7281695  3.407232
##    0.50    0.043754794  4.832380  0.7281668  3.402730
##    0.50    0.055410203  4.831926  0.7281538  3.397857
```

```
##    0.50    0.070170383  4.831984  0.7280757  3.392448
##    0.50    0.088862382  4.833136  0.7278771  3.386425
##    0.50    0.112533558  4.836118  0.7274809  3.380509
##    0.50    0.142510267  4.842461  0.7267166  3.377234
##    0.50    0.180472177  4.852353  0.7255827  3.377136
##    0.50    0.228546386  4.870951  0.7234989  3.382360
##    0.50    0.289426612  4.899792  0.7202916  3.400647
##    0.50    0.366524124  4.929304  0.7169687  3.423896
##    0.50    0.464158883  4.958964  0.7138715  3.449014
##    0.50    0.587801607  4.994496  0.7104918  3.476491
##    0.50    0.744380301  5.050397  0.7051602  3.513228
##    0.50    0.942668455  5.121662  0.6985085  3.569930
##    0.50    1.193776642  5.175958  0.6951532  3.612571
##    0.50    1.511775071  5.237230  0.6935732  3.662030
##    0.50    1.914481976  5.329118  0.6915465  3.744926
##    0.50    2.424462017  5.464350  0.6886591  3.850862
##    0.50    3.070290630  5.660907  0.6846162  3.992614
##    0.50    3.888155180  5.921404  0.6820803  4.185898
##    0.50    4.923882632  6.291648  0.6764095  4.465401
##    0.50    6.235507341  6.797329  0.6622056  4.842805
##    0.50    7.896522868  7.400063  0.6465935  5.293495
##    0.50   10.000000000  8.103378  0.6364385  5.834225
##    0.75    0.001000000  4.836690  0.7279229  3.421542
##    0.75    0.001266380  4.836690  0.7279229  3.421542
##    0.75    0.001603719  4.836690  0.7279229  3.421542
##    0.75    0.002030918  4.836690  0.7279229  3.421542
##    0.75    0.002571914  4.836690  0.7279229  3.421542
##    0.75    0.003257021  4.836690  0.7279229  3.421542
##    0.75    0.004124626  4.836690  0.7279229  3.421542
##    0.75    0.005223345  4.836690  0.7279229  3.421542
##    0.75    0.006614741  4.836690  0.7279229  3.421542
##    0.75    0.008376776  4.836710  0.7279226  3.421431
##    0.75    0.010608184  4.836068  0.7279748  3.419615
##    0.75    0.013433993  4.835350  0.7280287  3.417250
##    0.75    0.017012543  4.834629  0.7280729  3.414369
##    0.75    0.021544347  4.833932  0.7281066  3.410953
##    0.75    0.027283334  4.833425  0.7281114  3.407017
##    0.75    0.034551073  4.833008  0.7280994  3.402481
##    0.75    0.043754794  4.832856  0.7280528  3.397572
##    0.75    0.055410203  4.833426  0.7279187  3.392220
##    0.75    0.070170383  4.835443  0.7276272  3.386307
##    0.75    0.088862382  4.840003  0.7270548  3.381470
##    0.75    0.112533558  4.847571  0.7261622  3.380053
##    0.75    0.142510267  4.862229  0.7244927  3.383295
##    0.75    0.180472177  4.888362  0.7215788  3.395470
##    0.75    0.228546386  4.917226  0.7182479  3.417167
##    0.75    0.289426612  4.946246  0.7149746  3.440273
##    0.75    0.366524124  4.974950  0.7119673  3.465805
##    0.75    0.464158883  5.020260  0.7074320  3.496109
##    0.75    0.587801607  5.083766  0.7010340  3.543089
##    0.75    0.744380301  5.139083  0.6960326  3.588260
##    0.75    0.942668455  5.188635  0.6934659  3.629347
##    0.75    1.193776642  5.259491  0.6907996  3.693384
##    0.75    1.511775071  5.361902  0.6872161  3.781261
```

```
##   0.75    1.914481976  5.506020  0.6833882  3.888623
##   0.75    2.424462017  5.706964  0.6799755  4.035583
##   0.75    3.070290630  6.008052  0.6729193  4.260038
##   0.75    3.888155180  6.446079  0.6567268  4.599301
##   0.75    4.923882632  6.981752  0.6436352  4.985612
##   0.75    6.235507341  7.676483  0.6316435  5.497847
##   0.75    7.896522868  8.559988  0.5763886  6.199007
##   0.75   10.000000000  9.093140        NaN  6.656521
##   1.00    0.001000000  4.836615  0.7279357  3.421427
##   1.00    0.001266380  4.836615  0.7279357  3.421427
##   1.00    0.001603719  4.836615  0.7279357  3.421427
##   1.00    0.002030918  4.836615  0.7279357  3.421427
##   1.00    0.002571914  4.836615  0.7279357  3.421427
##   1.00    0.003257021  4.836615  0.7279357  3.421427
##   1.00    0.004124626  4.836615  0.7279357  3.421427
##   1.00    0.005223345  4.836615  0.7279357  3.421427
##   1.00    0.006614741  4.836603  0.7279371  3.421349
##   1.00    0.008376776  4.836203  0.7279680  3.419911
##   1.00    0.010608184  4.835522  0.7280182  3.417621
##   1.00    0.013433993  4.834848  0.7280593  3.414836
##   1.00    0.017012543  4.834214  0.7280879  3.411511
##   1.00    0.021544347  4.833748  0.7280907  3.407690
##   1.00    0.027283334  4.833427  0.7280702  3.403222
##   1.00    0.034551073  4.833368  0.7280155  3.398398
##   1.00    0.043754794  4.834073  0.7278690  3.393130
##   1.00    0.055410203  4.836418  0.7275414  3.387423
##   1.00    0.070170383  4.841459  0.7269127  3.382811
##   1.00    0.088862382  4.849269  0.7259970  3.382069
##   1.00    0.112533558  4.865842  0.7241078  3.386840
##   1.00    0.142510267  4.893700  0.7209779  3.400972
##   1.00    0.180472177  4.921924  0.7176848  3.422384
##   1.00    0.228546386  4.950280  0.7144407  3.445367
##   1.00    0.289426612  4.979325  0.7113398  3.471195
##   1.00    0.366524124  5.028411  0.7062696  3.503732
##   1.00    0.464158883  5.087574  0.7002001  3.548916
##   1.00    0.587801607  5.142487  0.6949330  3.594093
##   1.00    0.744380301  5.192871  0.6919565  3.637952
##   1.00    0.942668455  5.264424  0.6885141  3.704611
##   1.00    1.193776642  5.364412  0.6843656  3.788128
##   1.00    1.511775071  5.496356  0.6811964  3.889111
##   1.00    1.914481976  5.698927  0.6759074  4.040179
##   1.00    2.424462017  6.007488  0.6648269  4.271241
##   1.00    3.070290630  6.441553  0.6454123  4.608152
##   1.00    3.888155180  6.970209  0.6357967  4.974705
##   1.00    4.923882632  7.733097  0.6044262  5.530644
##   1.00    6.235507341  8.679876  0.5699586  6.299511
##   1.00    7.896522868  9.093140        NaN  6.656521
##   1.00   10.000000000  9.093140        NaN  6.656521
##
## RMSE was used to select the optimal model using the smallest value.
## The final values used for the model were alpha = 0.25 and lambda = 0.1125336.
```

```
glmnet_metrics
```

```
## # A tibble: 1 x 4
##   Model                 RMSE   MAE    R2
##   <chr>                <dbl> <dbl> <dbl>
## 1 Elastic Net (glmnet)  4.59  3.37 0.764
```

Coefficient (Interpretable Drivers)

```r
best <- glmnet_fit$bestTune
best
```

```
##    alpha    lambda
## 61  0.25 0.1125336
```

```r
final_glmnet <- glmnet_fit$finalModel
coef_mat <- coef(final_glmnet, s = best$lambda)

coef_df <- tibble(
  term = rownames(as.matrix(coef_mat)),
  estimate = as.numeric(as.matrix(coef_mat))
) %>%
  filter(term != "(Intercept)") %>%
  arrange(desc(abs(estimate)))

head(coef_df, 12)
```

```
## # A tibble: 12 x 2
##    term    estimate
##    <chr>      <dbl>
##  1 lstat     -3.89
##  2 rm         2.64
##  3 dis       -2.64
##  4 rad        2.17
##  5 ptratio   -2.00
##  6 nox       -1.70
##  7 tax       -1.46
##  8 black      0.912
##  9 zn         0.724
## 10 crim      -0.669
## 11 chas       0.612
## 12 indus     -0.198
```

Random Forest (Non-linear + Interactions)

```r
set.seed(42)
rf_fit <- train(
  medv ~ ., data = train,
  method = "rf",
  trControl = ctrl,
  tuneLength = 8
```

```
)

rf_pred <- predict(rf_fit, newdata = test)

rf_metrics <- tibble(
  Model = "Random Forest",
  RMSE = rmse(y_test, rf_pred),
  MAE  = mae(y_test, rf_pred),
  R2   = r2(y_test, rf_pred)
)

rf_fit
```

```
## Random Forest
##
## 407 samples
##  13 predictor
##
## No pre-processing
## Resampling: Cross-Validated (10 fold, repeated 3 times)
## Summary of sample sizes: 366, 367, 367, 366, 365, 367, ...
## Resampling results across tuning parameters:
##
##   mtry  RMSE      Rsquared   MAE
##    2    3.730540  0.8500965  2.504164
##    3    3.455975  0.8689108  2.336131
##    5    3.286445  0.8780783  2.237428
##    6    3.257238  0.8800641  2.218811
##    8    3.257554  0.8779761  2.221670
##    9    3.277066  0.8762192  2.233907
##   11    3.332730  0.8710574  2.267585
##   13    3.370363  0.8674099  2.290992
##
## RMSE was used to select the optimal model using the smallest value.
## The final value used for the model was mtry = 6.
```

```
rf_metrics
```

```
## # A tibble: 1 x 4
##   Model          RMSE   MAE    R2
##   <chr>         <dbl> <dbl> <dbl>
## 1 Random Forest  3.10  2.12 0.896
```

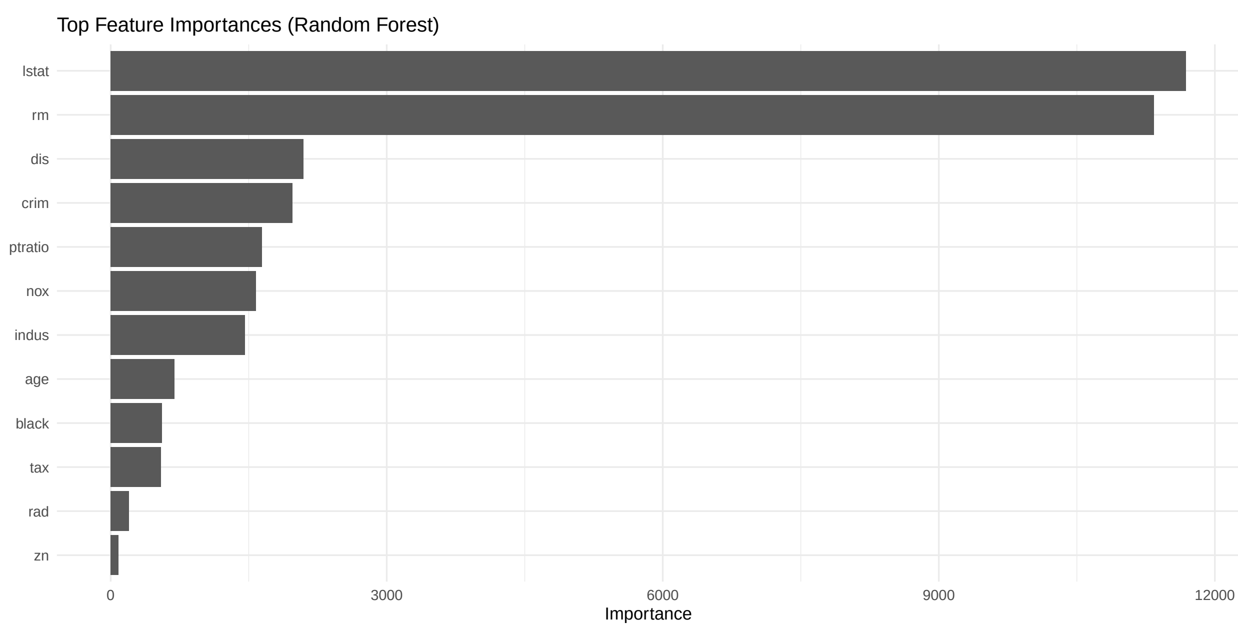
Feature Importane (Random Forest)

```
rf_final <- rf_fit$finalModel
imp <- randomForest::importance(rf_final)
imp_df <- tibble(
  Feature = rownames(imp),
  Importance = imp[,1]
) %>% arrange(desc(Importance))
```

```r
head(imp_df, 12)
```

```
## # A tibble: 12 x 2
##    Feature Importance
##    <chr>        <dbl>
##  1 lstat      11687.
##  2 rm         11335.
##  3 dis         2095.
##  4 crim        1977.
##  5 ptratio     1644.
##  6 nox         1578.
##  7 indus       1458.
##  8 age          690.
##  9 black        555.
## 10 tax          544.
## 11 rad          198.
## 12 zn            84.1
```

```r
ggplot(imp_df %>% slice_max(Importance, n=12),
       aes(reorder(Feature, Importance), Importance)) +
  geom_col() +
  coord_flip() +
  labs(title="Top Feature Importances (Random Forest)",
       x=NULL,
       y="Importance")
```



Evaluation

Compare Models on Test Set

```r
results <- bind_rows(baseline_metrics,
                     lm_metrics,
                     glmnet_metrics,
                     rf_metrics) %>%
  arrange(RMSE)

results
```

```
## # A tibble: 4 x 4
##   Model                  RMSE   MAE     R2
##   <chr>                 <dbl> <dbl>  <dbl>
## 1 Random Forest          3.10  2.12  0.896
## 2 Linear Regression      4.59  3.37  0.761
## 3 Elastic Net (glmnet)   4.59  3.37  0.764
## 4 Baseline (Train Mean)  9.32  6.60 NA
```

Prediction Quality Plot (Best Model)

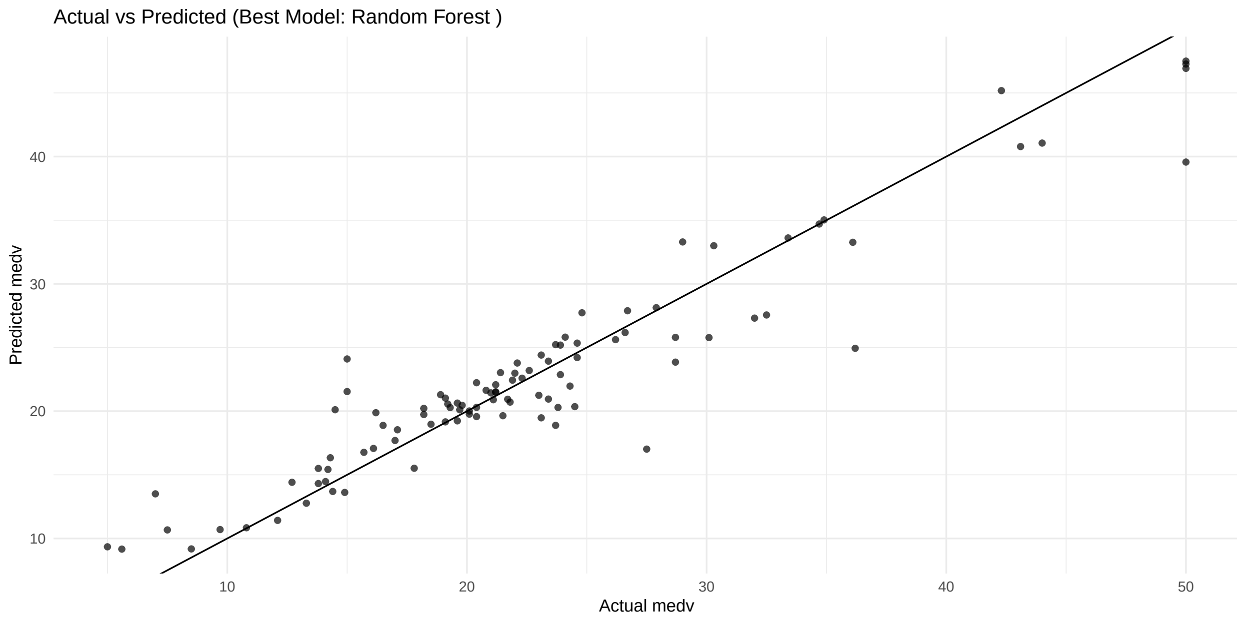
```r
best_model <- results$Model[1]
best_model
```

```
## [1] "Random Forest"
```

```r
best_pred <- switch(
  best_model,
  "Baseline (Train Mean)" = baseline_pred,
  "Linear Regression" = lm_pred,
  "Elastic Net (glmnet)" = glmnet_pred,
  "Random Forest" = rf_pred
)

plot_df <- tibble(
  actual = y_test,
  predicted = best_pred
)

ggplot(plot_df, aes(actual, predicted)) +
  geom_point(alpha = 0.7) +
  geom_abline(slope = 1, intercept = 0) +
  labs(
    title = paste("Actual vs Predicted (Best Model:", best_model, ")"),
    x = "Actual medv",
    y = "Predicted medv"
  )
```



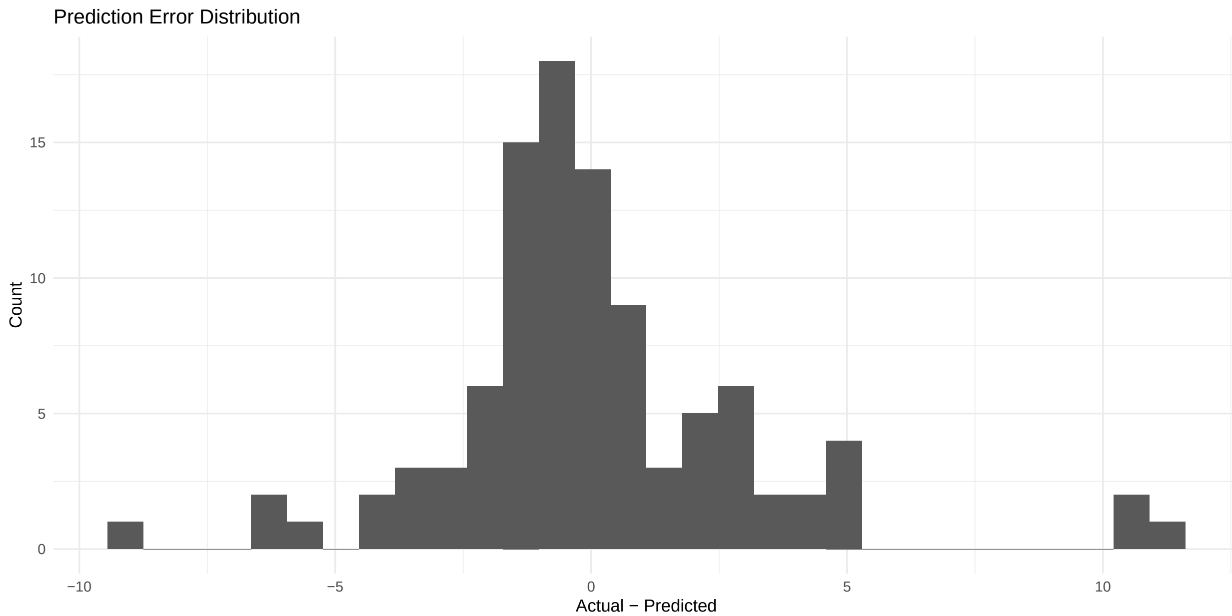
Error Analysis (Where the model misses)

```
plot_df <- plot_df %>%
  mutate(error = actual - predicted,
         abs_error = abs(error)) %>%
  arrange(desc(abs_error))

head(plot_df, 10)
```

```
## # A tibble: 10 x 4
##    actual predicted error abs_error
##     <dbl>     <dbl> <dbl>     <dbl>
##  1   36.2      24.9 11.3      11.3
##  2   27.5      17.0 10.5      10.5
##  3   50        39.6 10.4      10.4
##  4   15        24.1 -9.10      9.10
##  5   15        21.5 -6.54      6.54
##  6    7        13.5 -6.50      6.50
##  7   14.5      20.1 -5.61      5.61
##  8   32.5      27.6  4.94      4.94
##  9   28.7      23.9  4.85      4.85
## 10   23.7      18.9  4.82      4.82
```

```
ggplot(plot_df, aes(error)) +
  geom_histogram(bins = 30) +
  labs(title="Prediction Error Distribution",
       x="Actual - Predicted",
       y="Count")
```


Prediction Error Distribution
Count
15
10
5
0
−10
−5
0
5
10
Actual − Predicted

Deployment

```r
score_medv <- function(newdata, best_model_name = best_model) {
  # Ensure columns match training structure
  if (!all(names(train %>% select(-medv)) %in% names(newdata))) {
    stop("newdata must include all predictor columns used in training.")
  }
  newdata <- newdata %>% select(names(train %>% select(-medv)))

  pred <- switch(
    best_model_name,
    "Linear Regression" = predict(lm_fit, newdata = newdata),
    "Elastic Net (glmnet)" = {
      x_new <- predict(preproc, newdata)
      predict(glmnet_fit, newdata = as.matrix(x_new))
    },
    "Random Forest" = predict(rf_fit, newdata = newdata),
    rep(mean(y_train), nrow(newdata))
  )
  return(as.numeric(pred))
}

# Example: score the first 3 rows of the test set predictors
example_new <- test %>% select(-medv) %>% slice(1:3)
score_medv(example_new)
```

```
## [1] 34.70179 25.79722 18.88134
```

Conclusion

References